THE SPACE OF CONNECTED COMPONENTS OF A QUASI-COMPACT SPACE WHOSE COMPONENTS ARE INTERSECTIONS OF CLOPENS IS PROFINITE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of profinite spaces as Hausdorff quasi-compact totally disconnected. Invoked at the start of the counted proof (source line 4457). Reduces the assertion to the three properties.
- The statement that the image of a quasi-compact space is quasi-compact, and that the space of components is totally disconnected by construction. Invoked inside the counted proof (source lines 4458-4461). Gives quasi-compactness and total disconnectedness.
- The statement that a component is the intersection of the clopens containing it, together with the finite intersection criterion. Invoked inside the counted proof (source lines 4463-4468). Produces a clopen separating two components.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-connected-components: sources/stacks-topology/pinned-original-source.tex, source lines 875–875. Navigation pointer only: the counted text refers to the Stacks item lemma-connected-components, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted text refers to the Stacks item lemma-image-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-profinite: sources/stacks-topology/pinned-original-source.tex, source lines 4351–4351. Navigation pointer only: the counted text refers to the Stacks item lemma-profinite, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-space-connected-components: sources/stacks-topology/pinned-original-source.tex, source lines 1014–1014. Navigation pointer only: the counted text refers to the Stacks item lemma-space-connected-components, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. If X is quasi-compact and every connected component of X is the intersection of the open and closed subsets containing it, then $\pi_0(X)$ is a profinite space.*

Proof. We will use Lemma lemma-profinite to prove this. Since $\pi_0(X)$ is the image of a quasi-compact space it is quasi-compact (Lemma lemma-image-quasi-compact). It is totally disconnected by construction (Lemma lemma-space-connected-components). Let $C, D \subset X$ be distinct connected components of X. Write $C = \bigcap U_\alpha$ as the intersection of the open and closed subsets of X containing C. Any finite intersection of U_α's is another. Since $\bigcap U_\alpha \cap D = \emptyset$ we conclude that $U_\alpha \cap D = \emptyset$ for some α (use Lemmas lemma-connected-components, lemma-closed-in-quasi-compact and lemma-intersection-closed-in-quasi-compact) Since U_α is open and closed, it is the union of the connected components it contains, i.e., U_α is the inverse image of some open and closed subset $V_\alpha \subset \pi_0(X)$. This proves that the points corresponding to C and D are contained in disjoint open subsets, i.e., $\pi_0(X)$ is Hausdorff. □